Reduction in cardiovascular risk by sodium-bicarbonated mineral water in moderately hypercholesterolemic young adults.

The effects of drinking sodium-bicarbonated mineral water on cardiovascular risk in young men and women with moderate cardiovascular risk were studied. Eighteen young volunteers (total cholesterol levels >5.2 mmol/L) without any disease participated. The study consisted of two 8-week intervention periods. Subjects consumed, as supplement to their usual diet, 1 L/day control low mineral water, followed by 1 L/day bicarbonated mineral water (48 mmol/L sodium, 35 mmol/L bicarbonate and 17 mmol/L chloride). Determinations were performed at the end of the control water period and on Weeks 4 and 8 of the bicarbonated water period. Body weight, body mass index (BMI), blood pressure, dietary intake, total cholesterol, low-density lipoprotein (LDL) cholesterol, high-density lipoprotein (HDL) cholesterol, apolipoprotein (Apo) A-I, Apo B, triacylgycerols, glucose, insulin, adiponectin, high-sensitivity C-reactive protein (hs-CRP), soluble adhesion molecules [soluble intercellular adhesion molecule (sICAM) and soluble vascular adhesion molecule (sVCAM)], sodium and chloride urinary excretion, and urine pH were measured. Dietary intake, body weight and BMI showed no significant variations. Systolic blood pressure decreased significantly after 4 weeks of bicarbonated water consumption, without significant differences between Weeks 4 and 8. After bicarbonated water consumption, significant reductions in total cholesterol (by 6.3%; P=.012), LDL cholesterol (by 10%; P=.001), total/HDL cholesterol (P=.004), LDL/HDL cholesterol (P=.001) and Apo B (P=.017) were observed. Serum triacylglycerol, Apo A-I, sICAM-1, sVCAM-1 and hs-CRP levels did not change. Serum glucose values tended to decrease during the bicarbonated water intervention (P=.056), but insulin levels did not vary. This sodium-bicarbonated mineral water improves lipid profile in moderately hypercholesterolemic young men and women and could therefore be applied in dietary interventions to reduce cardiovascular risk.